3 Continuous Time Fourier Series (CTFS)

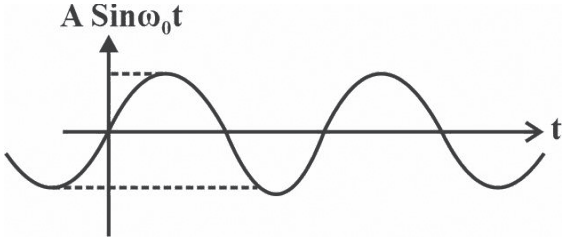
Objectives

Upon completion of this chapter, the user will be able to understand:

- Visualising signal as a vector
- Different representations of non-sinusoidal periodic signals as a weighted sum of mutually orthogonal, harmonically related sinusoids.
- Harmonic calculations
- Different properties
- Various types of symmetries and its effect on EFSC and TFSC
- Exact calculations of EFSC and TFSC
- Parseval's theorem
- Dirichlet's conditions
- Gibb's phenomenon
- Note of common mistakes usually committed.

Introduction

In the previous chapter, we learned different types of systems and applications of convolution operators and how one can obtain the response of a linear system to an arbitrary input by representing it in terms of basic signals. In this chapter, we will learn various representations of non-sinusoidal periodic signals as the weighted sum of mutually orthogonal, harmonically related sinusoids with the help of tools given by Jean Baptiste Joseph Fourier, with which we can understand the equivalent frequency domain characteristics of periodic signals and how to decompose periodic signals into their frequency components.



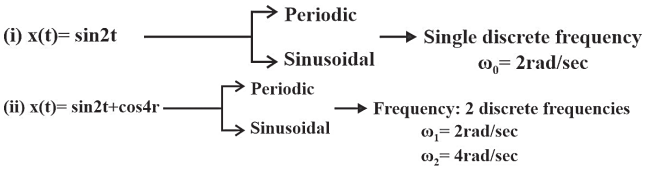
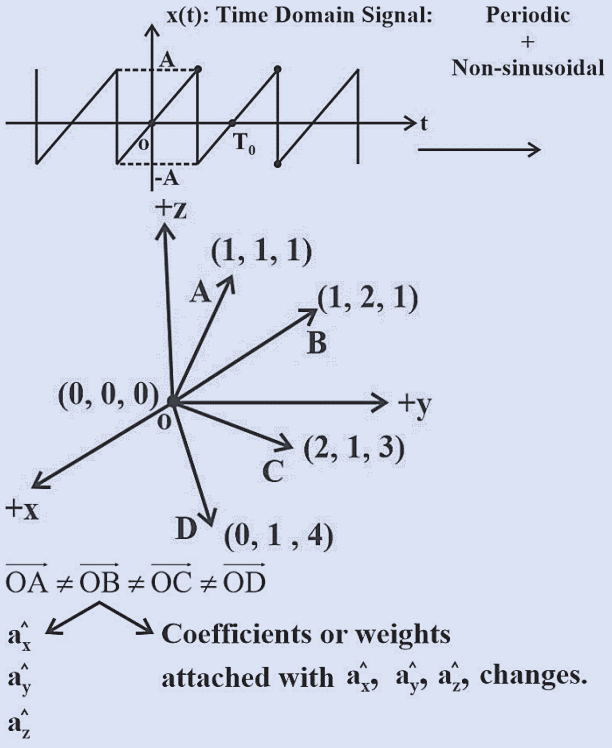


Fig. 3.1

Note: Time domain sinusoidal signal → Single discrete frequency



$\overrightarrow{OA} = 1a_{\hat{x}} + 1a_{\hat{y}} + 1a_{\hat{z}}$

$\overrightarrow{OB} = 1a_{\hat{x}} + 2a_{\hat{y}} + 1a_{\hat{z}}$

$\overrightarrow{OC} = 2a_{\hat{x}} + 1a_{\hat{y}} + 3a_{\hat{z}}$

$\overrightarrow{OD} = 0a_{\hat{x}} + 1a_{\hat{y}} + 4a_{\hat{z}}$

$S = \{\hat{a}_x, \ a_{\hat{y}}, \ \hat{a}_z\}$

$\widehat{a_i} \cdot \widehat{a_j} = 0 \quad \hat{i} \neq \hat{j}$

$\widehat{a_i} \cdot \widehat{a_j} = 1 \quad \hat{i} = \hat{j}$ → Inner product of vectors

$S = \{a_{\hat{x}}, a_{\hat{y}}, a_{\hat{z}}\}$ → "Complete set of mutually orthogonal unit vectors in 3-D space."

→ Any vector in 3-D space can be written as the weighted sum of mutually orthogonal unit vectors.

→ As vector changes, weight changes.

Case 1: $\vec{X} = \{\vec{X}_1, \vec{X}_2, \vec{X}_3, \ldots \vec{X}_n\}$ Set of column vectors.

Inner product: $[X_i], [X_j] = \begin{cases} 0 & i \neq j \\ \neq 0 & i = j \end{cases}$

X : set of orthogonal

↓

1 orthogonal

Case 2: $S = \{\vec{A}_1, \vec{A}_2, \vec{A}_3 \ldots \vec{A}_n\}$ Set of vectors

Inner product: $\vec{A}_i \cdot \vec{A}_j = \begin{cases} 0 & i \neq j \\ \neq 0 & i = j \end{cases}$

δ : set of orthogonal

↓

1 orthogonal

Case 3: $S = \{x_2(t), x_2(t) \ldots . x_n(t)\}$

Inner product: $\int_{-\infty}^{\infty} x_i(t) x_j^*(t) dt = \begin{cases} 0 & i \neq j \\ \neq 0 & i = j \end{cases}$

set of orthogonal

1 orthogonal

Case 4: $S = \{x_1, x_2, x_3 \quad \ldots x_n\}$

X: Random variable

Inner product: $E[X_i X_j] = \begin{cases} 0 & i \neq j \\ \neq 0 & i = j \end{cases}$

set of orthogonal

1 orthogonal

Case 1: Set of mutually orthogonal signals in the interval $(0, 2\pi / \omega_0)$

$S_1 = \{\cos m\omega_0 t\}_{m=0}^{\infty}$

$S_1 = \{1, \cos \omega_0 t, \cos 2\omega_0 t, \cos 3\omega_0 t \cdots\cdots\}$

Verification:

Similar:

$\rightarrow \int_0^{2\pi/\omega_0} 1(1) * dt = \frac{2\pi}{\omega_0} \neq 0$

$\rightarrow \int_0^{2\pi/\omega_0} \cos \omega_0 t (\cos \omega_0 t) * dt$

$\int_0^{2\pi/\omega_0} \cos^2 \omega_0 dt = \frac{\pi}{\omega_0} \neq 0$

Dissimilar:

$\int_0^{2\pi/\omega_0} \cos\omega_0 t \left(\cos 2\omega_0 t\right)^* dt = \int_0^{2\pi/\omega_0} \frac{1}{2}$

$\cos 3\omega_0 t \ dt + \int_0^{2\pi/\omega_0} \frac{1}{2} \cos \omega_0 t dt = 0$

→ Why not $\{\cos m\omega_0 t\}_{n=-\infty}^{n=\infty}$

$\{\cdots \cos(-2\omega_0 t), \cos(-\omega_0 t), 1, \quad \}$
$\{\cos \omega_0 t, \cos 2\omega_0 t \cdots \quad \}$

Case 2: Set of mutually orthogonal signals in $(0, \pi / \omega_0)$

$S_2 = \{\sin n\omega_0 t\}_{n=0}^{\infty}$

$S_2 = \{\sin \omega_0 t, \ \sin 2\omega_0 t, \ldots\ldots\}$

Verify:

Similar

$\int_0^{2\pi/\omega_0} \sin\omega_0 t \left(\sin \omega_2 t\right)^* dt = \pi / \omega_0 \neq 0$

Dissimilar:

$\int_0^{2\pi/\omega_0} \sin\omega_0 t \left(\sin 2\omega_0 t\right)^* dt = 0$

→Why not: $\{\sin n\omega_0 t\}_{n=0}^{\infty} = \{0, \sin \omega_0 t \cdot \sin 2\omega_0 t \ldots\}$

Similar: $\int_0^{2\pi/\omega_0} 0(0)^* dt = 0$

→ Why not

$\{\sin n\omega_0 t\}_{n=0}^{\infty} = \{\ldots - \sin \omega_0 t, \ 0, \ \sin \omega_0 t \ldots\}$

Dissimilar:

$\int_0^{2\pi/\omega_0} \left(\sin \omega_0 t\right)\left(-\sin \omega_0 t\right)^* dt = \frac{-\pi}{\omega_0} \neq 0$

Case 3: Complete set of mutually orthogonal, harmonically related sinusoids in $(0 - 2\pi / \omega_0)$.

$S = \{1, \cos n\omega_0 t, \sin n\omega_0 t\}_{n=1}^{\infty}$

$S = \{1, \cos \omega_0 t, \cos 2\omega_0 t \cdots, \sin \omega_0 t, \ \sin 2\omega_0 t \cdots\}$

Dissimilar:

$$\int_0^{2\pi\omega_0} \cos\omega_0 t \left(\sin 2\omega_0 t\right)^* dt = 0$$

1.

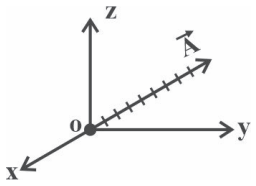


Fig. 3.2

$$\hat{S} = \left\{\hat{a}_x, a_y^{'}, a_z^{\hat{2}}\right\} \qquad \overrightarrow{OA} = l\hat{a}_x + m\hat{a}_y + n\hat{a}_z$$

→ as $\overrightarrow{OA}$ changes, l, m, n changes

2.

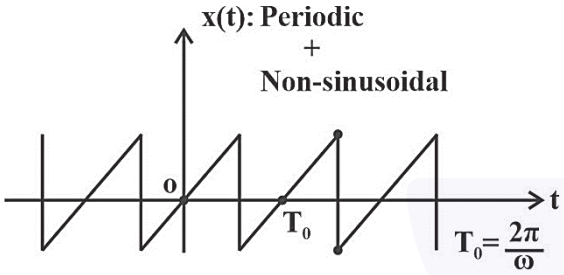


Fig. 3.3

$$S = \left\{1, \cos n\pi\omega_0 t, \sin m\omega_0 t\right\}_{m=L}^{\infty}$$

$$a_0 + a_1 \cos\omega_0 t + a_2 \cos 2\omega_0 t + \cdots$$
$$+b_1 \sin\omega_0 t + b_2 \sin 2\omega_0 t + \cdots$$

Fourier Series

1. Fourier series is the representation of time domain non-sinusoidal periodic signal as the weighted sum of mutually orthogonal, harmonically related sinusoids.
2. Fourier series provides the method to break any time domain non-sinusoidal periodic signal as the weighted sum of mutually orthogonal, harmonically related sinusoids.

Purpose: "To calculate the frequency of time domain non-sinusoidal periodic signal."

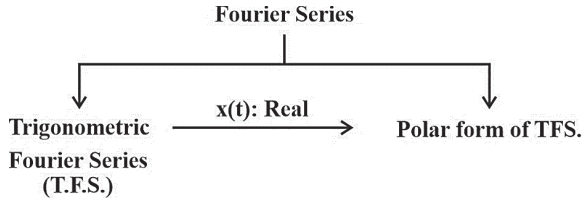


Fig. 3.4

TFS:

$$T_0 = \frac{2\pi}{\omega_0}$$

$$\omega_0 = HCF\left(\omega_0, 2\omega_0 \ldots\right) = \omega_0$$

$$T_0 = 2\pi / \omega_0$$

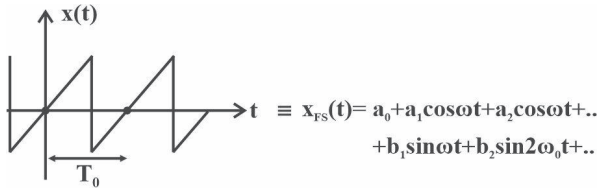


Fig. 3.5

$$x_{Fs}(t) = a_0 + \sum_{n=1}^{\infty}\left\{a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right\}$$

$a_0, a_n, b_n \rightarrow$ Trigonometric Fourier series coefficients (T.F.S.C)

TFS:

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left\{a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right\}$$

$$\rightarrow a_0 = \frac{1}{T_0}\int_{\langle T_0\rangle} x(t)dt = \frac{Area\ in\ 1\ T_0}{T_0}$$

→ a_0 : average value or dc value of x(t)

$$a_n = \frac{2}{T_0}\int_{\langle T_0\rangle} x\left(t\right)\cos n\omega_0 t\ dt \rightarrow n \geq 1 \rightarrow \boxed{a_{-n} = a_n}$$

$$b_n = \frac{2}{T_0}\int_{\langle T_0\rangle} x\left(t\right)\sin n\omega_0 t\ dt \rightarrow n \geq 1 \rightarrow \boxed{b_{-n} = -b_n}$$

x(t)	a_0	a_n	b_n
Real	Real	Real	Real
Purely 1	I	I	I
Complex	Complex	Complex	Complex

Table 3.1

Polar form of TFS:

$$\rightarrow x\left(t\right) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

→ if $x\left(t\right)$ is Real → $a_0, a_n, b_n \rightarrow$ Real

Let

$$a_n = r_n \cos\phi_n \qquad r_n = \sqrt{a_n^2 + b_n^2}$$
$$b_n = r_n \sin\phi_n \qquad \phi_n = \tan^{-1}\left(\frac{b_n}{a_n}\right)$$

$$r_n = \sqrt{a_n^2 + b_n^2} = \begin{cases} r_0 = a_0 & n = 0 \\ r_n : & n \geq 1 \end{cases}$$

$$\phi_n = \tan^{-1}\frac{b_n}{a_n} = \begin{cases} 0° & n = 0 \\ \phi_n : & n \geq 1 \end{cases}$$

$$x(t) = r_0 + \sum_{n=1}^{\infty} r_n \cos\left(n\omega_0 t - \phi_n\right)$$

$$x(t) = \sum_{n=0}^{\infty} r_n \cos\left(n\omega_0 t - \phi_n\right)$$

Use of polar form:

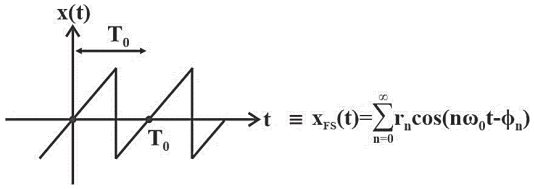


Fig. 3.6

$$\equiv r_0 + r_1 \cos\left(\omega_0 t - \phi_1\right) + r_2 \cos\left(2\omega_0 t - \phi_2\right) + \cdots\cdots$$

r_0 : DC component of time domain, non-sinusoidal P signal x(t):

→ Frequency of dc component: OHz

→ Amplitude = r_0

→ Power = r_0^2

→ RMS = r_0

$r_1 \cos\left(\omega_0 t - \phi_1\right)$: I harmonic or fundamental harmonic of time domain non-sinusoids P signal x(t):

Frequency → ω_0 / f_0

Amplitude → $r_1 = \sqrt{a_1^2 + b_1^2}$

Power → $\frac{r_1^2}{2}$

RMS → $\frac{r_1}{\sqrt{2}}$

$r_k \cos\left(k\omega_0 t - \phi_k\right)$: k^{th} harmonic of x(t)

Frequency → $k\omega_0 / kf_0$

Amplitude → $r_k = \sqrt{ak^2 + bk^2}$

Power → $\frac{rk^2}{2}$

RMS → $\frac{r_k}{\sqrt{2}}$

Power of time domain, non-sinusoidal P signal:

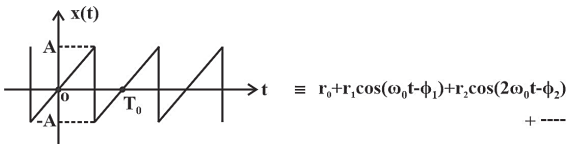


Fig. 3.7

$$\overline{x^2(t)} = \frac{A^2}{3} \qquad \overline{x_{FS(1)}^2} = r_0^2 + \frac{r_1^2}{2} + \frac{r_2^2}{2} + \cdots$$

$$\frac{A^2}{3} = r_0^2 + \sum_{n=1}^{\infty} \frac{r_n^2}{2}$$

$$r_0^2 + \sum_{n=1}^{K} \frac{r_n^2}{2} = m\%\left\{\overline{x^2(t)}\right\} \rightarrow \text{Total Abs power:}$$

Absent harmonics calculation:

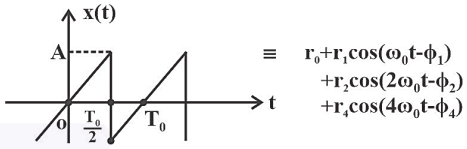


Fig. 3.8

$r_n \rightarrow \left(r_n\right)_{n=1}$: $r_1 = 0$

S.1→ $\omega_0 T_0$

S.2→ a_0, a_n, b_n

S.3→ r_0
r_n

S.4→ Absent frequency will be those harmonics for which $r_n = 0$.

$r_k = 0 \rightarrow k^{th}$ harmonic is absent

Example 3.1:

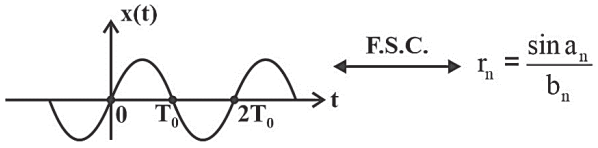


Fig. 3.9

Calculate absent frequency in x(t) = ?

Solution:

Harmonic absent: $r_n = 0$

$$\frac{\sin\frac{\pi}{2}n}{n} = \frac{\pi}{2}\text{sinc}\left(\frac{n}{2}\right) \xrightarrow[over]{\text{zerocross}}$$

$$\frac{n}{2} = k \quad k \in I \quad k \neq 0$$

$n = 2k \quad k \neq 0$

$n = 2, 4, 6, 8, 10....$

Harmonics absent: 2, 4, 6, 8 ...

Dc component: present

$$x(t) \xrightarrow{\text{Freq}^n \text{ present}} 0,\ f_0,\ 3f_0,\ 5f_0$$

S_4: Complete set of mutually orthogonal, harmonically related, complex exponentials in $2\pi / \omega_0$.

$$S = \left\{e^{jn\omega_0 t}\right\}_{n=-\infty}^{n=\infty} = \left\{\cdots e^{-j\omega_0 t}, 1, e^{+j\omega_0 t} \cdots\right\}$$

Dissimilar:

$$\int_0^{2\pi/\omega_0} e^{-j\omega_0 t}\left(e^{j\omega_0 t}\right)^* dt = 0$$

Similar:

$$\int_0^{2\pi/\omega_0} e^{j\omega_0 t}\left(e^{j\omega_0 t}\right)^* dt = \frac{2\pi}{\omega_0} \neq 0$$

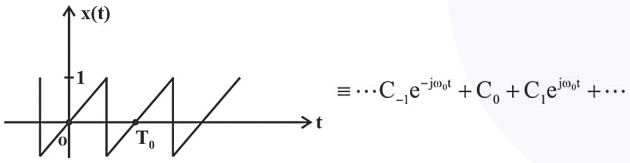


Fig. 3.10

$\equiv \sum_{\eta=-\infty}^{\infty} C_n e^{jn\omega_0 t} \rightarrow$ Complex or exponential FS

$C_n \rightarrow$ Complex or exponential FSC.

Complex FS $\rightarrow x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$

$$\begin{cases} c_0 = a_0 = r_0 = \dfrac{1}{T_0}\int_{\langle T_0\rangle} x(t)dt \\ c_n = \dfrac{1}{T_0}\int_{T_0} x(t)e^{-jn\omega_0 t}dt \end{cases}$$

Relation between a_0, a_n, b_n and c_0, c_n

$$C_n = \frac{1}{T_0}\int_{\langle T_0\rangle} x(t)e^{-jn\omega_0 t}dt$$

$$C_n = \frac{1}{T_0}\int_{\langle T_0\rangle} x(t)\cos n\omega_0 t dt - j\frac{1}{10}\int_{\langle T_0\rangle} x(t)$$

$\sin n\omega_0 t dt$

$$C_n = \frac{a_n}{2} - j\frac{b_n}{2}$$

$C_0 = a_0$

$$C_n = \frac{a_n}{2} - j\frac{b_n}{2}$$

2.

$$C_{-n} = \frac{a_{-n}}{2} - j\frac{b_{-n}}{2}$$

$$C_{-n} = \frac{a_n}{2} + j\frac{b_n}{2}$$

$C_{-n} = C_n^* \rightarrow$ only when x(t) is real $\rightarrow a_n, b_n, a_0$: Real

$$C_n = \frac{a_n}{2} - j\frac{b_n}{2}$$

$$|C_n| = \frac{\sqrt{a_n^2 + b_n^2}}{2} \xrightarrow{x(t):\text{Real}} |C_n| = \frac{r_n}{2}$$

Alternate example: $\rightarrow k^{th}$ harmonic

$r_k \cos\left(k\omega_0 t - \phi_k\right)$

1. Amp: $r_k = 2|C_k|$
2. RMS: $\dfrac{r_k}{\sqrt{2}} = 2|C_k|$
3. Power: $\dfrac{r_k^2}{2} = 2|C_k|^2$

$\rightarrow$ Absent frequency: $C_k = |C_k| e^{jhC_k}$

$\rightarrow$ If $C_k = 0 \rightarrow |C_k| = 0 \rightarrow \dfrac{r_k}{2} = 0 \rightarrow k^{th}$ harmonic absent

Numerical in Fourier series:

Type 1: Validity of T.F.S

Method:

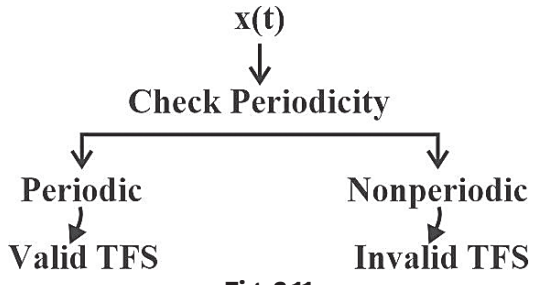


Fig. 3.11

3. $x(t) = 4\cos\left(\dfrac{\pi}{10}t\right) - 5\sin\left(10t - 15°\right)$, Is it a valid $T.F.S = ?$

Solution:

$T_1 = \frac{2\pi}{\pi/10} = 20 \qquad T_2 = \frac{2\pi}{10} = \frac{\pi}{5}$

$\frac{T_1}{T_2} \rightarrow IRR \;\; x(t) : NP \rightarrow$ Invalid $F.S.$

Type 2: Calculation of T.F.S.C. from T.F.S.

S.1→ $\omega_0 = HCF(\omega_1, \omega_2 \cdots\cdots) = \frac{HCFM^r}{LCMD^r}$

S.2→ Harmonics calculation:

$\omega_1 = k_1\omega_0$

$\omega_2 = k_2\omega_0$

S3→ $x(t) = a_0 + \sum_{\eta=L}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t \rightarrow$ Compare

Type 3: Calculation of complex FSC from the combination of sinusoids:

S.1→ $\omega_0 = HCF(\omega_1, \omega_2 \ldots)$

S.2→ Sinusoidal exponential in terms of exponentials

S.3→ $x(t) = \sum_{\eta=-\infty}^{\infty} C_n e^{jn\omega_0 t} \rightarrow$ Compare

4. Calculate EFSCa) $x(t) = \cos t$

S.1→ $\omega_0 = 1$

S.2→ $x(t) = \frac{e^{jt}}{2} + \frac{e^{-jt}}{2}$

S.3→ $x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} C_n e^{jnt}$

$= C_1 e^{jt} + C_{-1} e^{-jt}$

$C_1 = \frac{1}{2}$

$C_{-1} = \frac{1}{2}$

Note:

1. $C_n = \begin{cases} \frac{1}{2} & n = 1 \\ \frac{1}{2} & n = -1 \end{cases}$

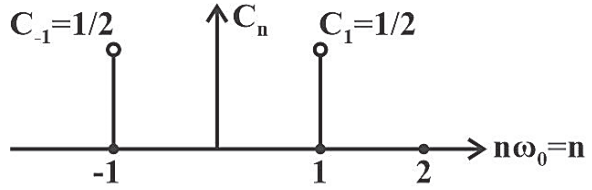
2. $C_n = \frac{1}{2}\delta(n+1) + \frac{1}{2}\delta(n-1)$

3. $C_1 = \frac{1}{2}, \quad C_{-1} = \frac{1}{2}$

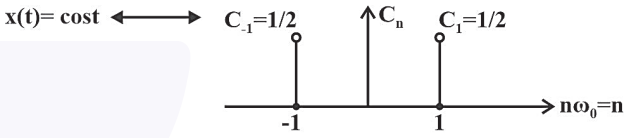
Plot of C_n: $C_n = \frac{1}{T_0}\int_{\langle T_0 \rangle} x(t)e^{-jn\omega_0 t}dt = f(n\omega_0)$

$x(t) = \cos t \xleftrightarrow{F.S.C.} C_n = \begin{cases} \frac{1}{2}: & n = 1 \\ \frac{1}{2}: & n = -1 \end{cases}$

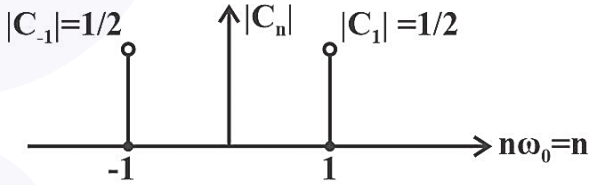
$C_n = |C_n| e^{jhC_n}$



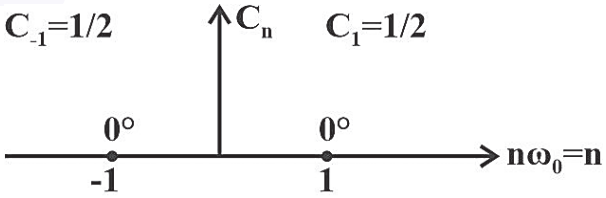
Amplitude spectrum



Amplitude spectrum



Magnitude specturm



Phase spectrum

Fig. 3.12

b) $x(t) = \cos 2t$

S.1→ $\omega_0 = 2$

S.2→ $x(t) = \frac{e^{j2t}}{2} + \frac{e^{-j2t}}{2}$

S.3→ $x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn(2)t} = C_1 e^{j2t} + C_{-1} e^{-j2t}$

$$\rightarrow C_n = \begin{cases} \frac{1}{2} & n = 1 \\ \frac{1}{2} & n = -1 \end{cases}$$

$$\rightarrow C_n = \frac{1}{2}\delta(n+1) + \frac{1}{2}\delta(n-1)$$

$$\rightarrow C_1 = \frac{1}{2}$$

$$C_{-1} = \frac{1}{2}$$

Plot:

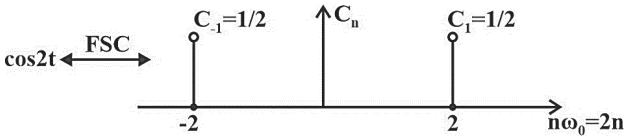


Fig. 3.13

c) $x(t) = \cos \omega_0 t$

S.1→ ω_0

S.2→ $x(t) = \frac{e^{j\omega_0 t}}{2} + \frac{e^{-j\omega_0 t}}{2}$

S.3→ $x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} = C_1 e^{j\omega_0 t} + C_{-1} e^{-j\omega_0 t}$

$$C_n = \begin{cases} \frac{1}{2} & n = 1 \\ \frac{1}{2} & n = -1 \end{cases}$$

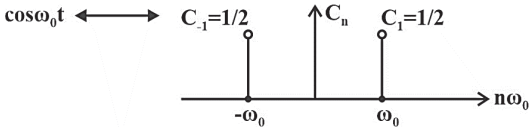


Fig. 3.14

Conclusion:

$\cos t \longleftrightarrow C_n = \frac{1}{2}$ $\quad n = \pm 1 \rightarrow C_n$ vs n

$\cos \omega_0 t \longleftrightarrow C_n = \frac{1}{2}$ $\quad n = \pm 1 \rightarrow C_n$ vs $n\omega_0$

$\cos k\omega_0 t \longleftrightarrow C_n = \frac{1}{2}$ $\quad n = \pm 1 \rightarrow C_n$ vs $kn\omega_0$

d) $x(t) = \sin t$

S.1→ $\omega_0 = 1$

S.2→ $x(t) = \frac{e^{jt}}{2j} - \frac{1}{2j}e^{-jt}$

S.3→ $x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jnt} = C_1 e^{jt} + C_{-1} e^{-jt}$

$$C_n = \begin{cases} \frac{1}{2j} & n = 1 \\ \frac{1}{2j} & n = -1 \end{cases}$$

OR

$$C_n = \frac{1}{2j}\delta(n-1) - \frac{1}{2j}\delta(n+1)$$

OR $C_1 = 1/2j$

$C_{-1} = -1/2j$

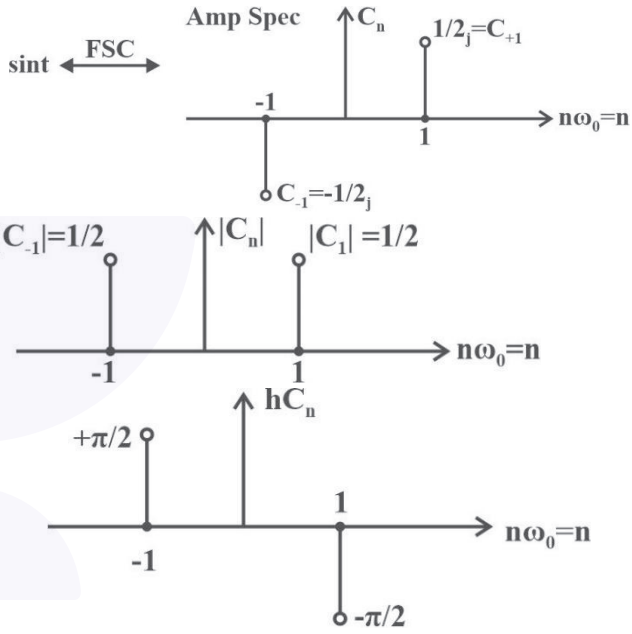


Fig. 3.15

$\frac{1}{2_i} = \frac{1}{2}e^{-j\pi/2}$

$\frac{-1}{2_i} = \frac{1}{2}e^{j\pi/2}$

e) $x(t) = \sin 2t \rightarrow$??

Conclusion:

$$\sin t \longleftrightarrow C_n = \begin{cases} \frac{1}{2i} & n = 1 \\ \frac{-1}{2i} & n = -1 \end{cases} : C_n \text{ vs n}$$

$$\sin \omega_0 t \longleftrightarrow C_n = \begin{cases} \frac{1}{2i} & n = 1 \\ \frac{-1}{2i} & n = -1 \end{cases} : C_n \text{ vs } n\omega_0$$

$$\sin k\omega_0 t \longleftrightarrow C_n = \begin{cases} \frac{1}{2i} & n=1 \\ \frac{-1}{2i} & n=-1 \end{cases} : C_n \text{ vs } n(k\omega_0)$$

Type 4: Properties of complex FSC

Linearity: Valid only when both signals have the same time period.

$$\omega_0 : x_1(t) \xleftrightarrow{FSC} C_n : C_n \text{ vs } n\omega_0$$

$$\omega_0 : x_2(t) \xleftrightarrow{FSC} d_n : d_n \text{ vs } n\omega_0$$

$$\omega_0 : g(t) = Ax_1(t) + Bx_2(t) \longleftrightarrow g_n$$
$$= AC_n + Bd_n : \ g_n \text{ vs } n\omega_0$$

Time shifting:

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 (t-t_0)}$$

$$x(t-t_0) = \sum_{n=-\infty}^{\infty} \left(C_n e^{-jn\omega_0 t_0}\right) e^{jn\omega_0 t}$$

$$\omega_0 : \quad x(t) \xleftrightarrow{FSC} C_n : \quad C_n \text{ vs } n\omega_0$$

$$\omega_0 : \quad g(t) = x(t-t_0) \longleftrightarrow g_n$$
$$= C_n e^{-jn\omega_0 t_0} : \quad g_n \text{ vs } n\omega_0$$

$$\omega_0 : \quad g(t) = x(t+t_0) \longleftrightarrow g_n$$
$$= C_n e^{jn\omega_0 t_0} : g_n \text{ vs } n\omega_0$$

Note:

$$\begin{cases} \omega_1 : x(t) \longleftrightarrow C_n : C_n \text{ vs } n\omega_0 \\ \omega_1 : x(t-t_0) \longleftrightarrow g_n = C_n e^{-jn\omega_0 t_0} : g_n \text{ vs } n\omega_1 \end{cases}$$

$$\begin{cases} 2\omega_0 : x(t) \longleftrightarrow C_n : C_n \text{ vs } n(2\omega_0) \\ 2\omega_0 : x(t-t_0) \longleftrightarrow g_n = C_n e^{-jn\omega_0 t_0} : \\ g_n \text{ vs } n(2\omega_0) \end{cases}$$

$$\omega_0 : \quad x(t) \longleftrightarrow C_n = |C_n| e^{j\angle C_n}$$

$$\omega_0 : g(t) = x(t \pm t_0) \longleftrightarrow g_n = e^{\pm jn\omega_0 t_0} C_n$$

$$g_n = e^{\pm jn\omega_0 t_0} |C_n| e^{j\angle C_n}$$

$$g_n = |C_n| e^{j[\angle C_n \mp n\omega_0 t_0]}$$

$$|g_n| = |C_n| \quad \angle g_n = \angle C_n \pm n\omega_0 t_0 ;$$

Time shifting properties does not change the magnitude spectrum.

5. $x(t) = \sin(2t - \pi/3)$, C.F.S.C.

$$\omega_0 = 2 : \sin 2t \longleftrightarrow C_n = \begin{cases} \frac{1}{2j} & n=1 \\ \frac{-1}{2j} & n=-1 \end{cases} \quad C_n \text{ vs } n\omega_0$$

$$\omega_0 = 2 : \sin 2(t + \pi/6) \longleftrightarrow g_n$$
$$= C_n e^{jn\omega_0 t_0} = C_n e^{jn\times 2\times \pi/6}$$

$$g_n = C_n e^{jn\pi/3} \rightarrow \begin{matrix} g_1 = C_1 e^{j\pi/3} \\ g_{-t} = C_{-1} e^{-j\pi/3} \end{matrix}$$

Time reversal:

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$x(-t) = \sum_{n=-\infty}^{\infty} C_n e^{-jn\omega_0 t}$$

$$x(-t) = \sum_{p=+\infty}^{-\infty} C_{-p} e^{jp\omega_0 t}$$

$$x(-t) = \sum_{p=-\infty}^{\infty} C_{-p} e^{jp\omega_0 t}$$

$$\omega_0 : x(t) \longleftrightarrow C_n : C_n \text{ vs } n\omega_0$$

$$\omega_0 : x(-t) \longleftrightarrow g_n = C_{-n} : g_n \text{ vs } n\omega_0$$

$x(t)$		C_n
Even	$\longleftrightarrow$	Even
Odd	$\longleftrightarrow$	Odd
Neno	$\longleftrightarrow$	Neno

6.

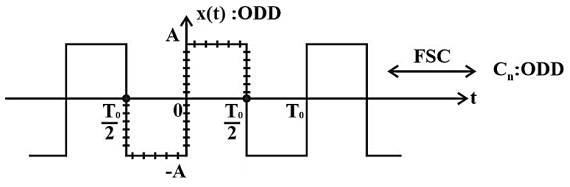
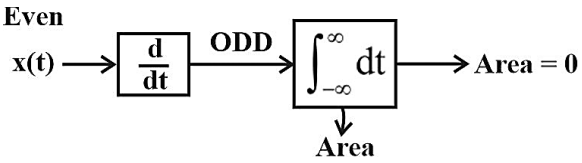


Fig. 3.16

$$\sum_{n=-\infty}^{\infty} C_n = 0$$



Time scaling:

$\omega_0 : \quad x(t) \longleftrightarrow C_n : C_n \text{ vs } n\omega_0$

$(a\omega_0) : g(t) = x(at) \longleftrightarrow g_n = C_n :$
$g_n \text{ vs } n(a\omega_0)$

$x(t) = \sum C_n e^{jn\omega_0 t} \rightarrow C_n \text{ vs } n\omega_0$

$x(at) = \sum C_n e^{jn(a\omega_0)t} \rightarrow C_n \text{ vs } n(a\omega_0)$

7. $x(t) \xleftrightarrow{FSC} C_n$

$x(-4t+1) \xleftrightarrow{FSC} g_n = ?$

Method 1:

$\omega_0 : x(t) \longleftrightarrow C_n : C_n \text{ vs } n\omega_0$

$\omega_0 : x(t+1) \longleftrightarrow C_n e^{jn\omega_0 t_0} = C_n e^{jn\omega_0}$
$\Rightarrow C_n e^{j\angle n\omega_0} \text{ vs } nw_0$

$4\omega_0 : x(4t+1) \longleftrightarrow C_n e^{jn\omega_0}$
$\Rightarrow C_n e^{j\angle n\omega_0} \text{ vs } n(4w_0)$

$4\omega_0 : x(-4t+1) \longleftrightarrow C_n e^{-jn\omega_0}$
$\Rightarrow C_n e^{-j\angle n\omega_0} \text{ vs } n(4w_0)$

Method 2:

$\omega_0 : x(t) \longleftrightarrow C_n : C_n \text{ vs } n\omega_0$

$4\omega_0 : x(4t) \longleftrightarrow C_n : C_n \text{ vs } (4n\omega_0)$

$4\omega_0 : x\left(4\left(t+\frac{1}{4}\right)\right) \longleftrightarrow C_n e^{jn[4\omega_0]\times\frac{1}{4}}$

$\Rightarrow C_n e^{jn\omega_0} \text{ vs } 4n\omega_0$

$4\omega_0 : x(-4t+1) \longleftrightarrow C_{-n} e^{jn\omega_0}$

$\Rightarrow C_{-n} e^{jn\omega_0} \text{ vs } 4n\omega_0$

Complex conjugate:

$\omega_0 : x(t) \longleftrightarrow C_n : C_n \text{ vs } n\omega_0$

$\omega_0 : x^*(t) \longleftrightarrow C^*_{-n} : C^*_{-n} \text{ vs } n\omega_0$

$x(t) = \sum C_n e^{jn\omega_0 t}$

$x^*(t) = \sum C^*_n e^{-jn\omega_0 t}$

$x^*(t) = \sum C^*_{-p} e^{jp\omega_0 t}$

Case 1:

x(t): Real

$x(t) = x*(t)$

$FSC\{x(t)\} = FSC\{x^*(t)\}$

$C_n = C^*_{-n} \rightarrow f(n\omega_0) = f^*(-n\omega_0) \rightarrow$ Conjugate symmetric

Case 2:

x(t): Imaginary

$x(t) = -x^*(t)$

FSC on both side

$C_n = -C^*_{-n} \rightarrow$ Conjugate anti symmetric

Case 3: x(t): conjugate symmetric

$x(t) = x^*(-t)$

FSC on both

$C_n = C^*_n \rightarrow$ Real.

$x^*(t) \longleftrightarrow C^*_{-n}$

$x^*(-t) \longleftrightarrow C^*_{-n}$

Case 4: x(t): Conjugate anti symmetric

$x(t) = -x^*(-t)$

$C_n = -C^*_n \rightarrow$ Imaginary

Important:

$x(t)$		C_n :
Real	↔	C.S.
Imag	↔	C.A.S
C.S.	↔	Real
C.A.S	↔	Imag

Case 5: x(t): Real + Even

$$\begin{matrix} Real \\ Even \end{matrix} + \begin{matrix} x(t) = x^*(t) \Leftrightarrow C_n = C^*_{-n} \\ x(t) = x(-t) \Leftrightarrow C_n = C_{-n} \end{matrix} + \quad C_n = C^*_n$$

→Real

Case 6: x(t): Real + Odd

$$\text{Real}: \quad x(t) = x^*(t) \Leftrightarrow C_n = C^*_{-n} \quad + \quad C_n = C^*_n$$
$$\text{Even}: \quad x(t) = x(-t) \Leftrightarrow C_n = -C_{-n}$$

→Imaginary

$x(t)$	C_n
R E	R E
R 0	I O
I E	I E
I 0	R 0

Table 1

$x(t)$		C_n
Even	↔	*Even*
Odd	↔	*Odd*
Neno	↔	Neno

Table 2

$x(t)$		C_n
Real	↔	*C.S.*
Imag	↔	*C.A.S.*
C.S.	↔	Real
C.A.S	↔	Imag.

Table 3

$x(t)$		C_n
R E	↔	*R E*
R O	↔	*I O*
I E	↔	I E
I O	↔	*R O*

8. Calculate FSC of following:

a) $\text{Re}\{x(t)\}$ d) $ODD\{x(t)\}$

b) $\text{Im}\{x(t)\}$ e) $CS\{x(t)\}$

c) $Ev\{x(t)\}$ f) $CAS\{x(t)\}$

Solution:

a) $\text{Re}\{x(t)\} = \dfrac{x(t)+x^*(t)}{2} \longleftrightarrow \dfrac{C_n + C^*_{-n}}{2}$

$= C_0\, S_0\,\{C_n\}$

b) $\text{Im}\{x(t)\} = \dfrac{x(t)-x^*(t)}{2j} \longleftrightarrow \dfrac{C_n - C^*_{-n}}{2i}$

$= \dfrac{1}{j}\; C_0\, A_0 S_0\,\{C_n\}$

c) $Ev\{x(t)\} = \dfrac{x(t)+x(-t)}{2} \longleftrightarrow \dfrac{C_n + C_{-n}}{2}$

$= Ev\{C_n\}$

d) $Odd\{x(t)\} = \dfrac{x(t)-x(-t)}{2} \longleftrightarrow \dfrac{C_n - C_{-n}}{2}$

$= odd\{C_n\}$

e) $C_0 S_0\{x(t)\} = \dfrac{x(t)+x^*(-t)}{2}$

$\longleftrightarrow \left\{\dfrac{C_n + C^*_{-n}}{2}\right\} = \text{Re}\{C_n\}$

f) $C_0 A_0 S_0\{x(t)\} = \dfrac{x(t)-x^*(-t)}{2} \longleftrightarrow$

$\dfrac{C_n + C^*_{-n}}{2} = j\,\text{Im}\{C_n\}$

9. Let x(t) is real calculate FSC $Ev\{x(t)\} = ?$

$$Ev\{x(t)\} = \frac{x(t)+x(-t)}{2} \leftrightarrow \frac{C_n + C_{-n}}{2}$$

Solution: $= \dfrac{C_n + C^*_n}{2} = \text{Re}\{C_n\}$

Important observation:

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$x(t) = \{C_0\} + \left[C_1 e^{j\omega_0 t} + C_{-1} e^{-j\omega_0 t}\right]$$
$$+ \left[C_2 e^{2j\omega_0 t} + C_{-2} e^{-j2\omega_0 t}\right] + \cdots\cdots$$

1. If $C_k = C^*_{-k}$: k^{th} harmonic will be real.
2. If $C_k = -C^*_{-k}$: k^{th} harmonic will be imaginary.
3. $C_K \neq C^*_{-k}$ & $C_K \neq -C^*_{-k}$: k^{th} harmonic will be complex.

3.1

$C_1 = 1 + 2i$
$C_{-1} = 1 - 2i$ $\quad C_1 = C^*_{-1}$

$= C_1 e^{j\omega_0 t} + C_{-1} e^{-j\omega_0 t}$

$= C_1 e^{j\omega_0 t} + C^*_1 e^{-j\omega_0 t}$

$= C_1 e^{j\omega_0 t} + \left(C_1 e^{j\omega_0 t}\right)^* = 2\,\text{Re}\left\{C_1 e^{j\omega_0 t}\right\}$

$= 2\,\text{Re}\left[(1+2i)\left(\cos\omega_0 t + j\sin\omega_0 t\right)\right] = \text{Real}$

Multiplication with complex exponential:

$\omega_0 : x(t) \longleftrightarrow C_n : C_n vs\ n\omega_0$

$\omega_0 : g(t) = e^{jm\omega_0 t} x(t) \longleftrightarrow g_n = C_{n-m} : g_n vs\ n\omega_0$

$\omega_0 : g(t) = e^{-jm\omega_0 t} x(t) \longleftrightarrow g_n = C_{n+m} : g_n vs\ n\omega_0$

10. $x(t) = e^{j2t}\cos t \longleftrightarrow C_n = ?$

Solution:

$\omega_2 = 1: \quad \cos t \longleftrightarrow C_n$

$$= \begin{cases} \frac{1}{2} & n = 1 \\ \frac{1}{2} & n = -1. \end{cases} : C_n vs\ n\omega_0$$

$w_0 = 1 : e^{j2t}\cos t \longleftrightarrow g_n$

$= C_{n-2} \quad g_n\ vs\ n\omega_0$

$g_3 = C_1 = 1/2 :$

$g_1 = C_{-1} = \frac{1}{2}$

$$g_n = \begin{cases} \frac{1}{2} & n = 1 \\ \frac{1}{2} & n = 3 \end{cases}$$

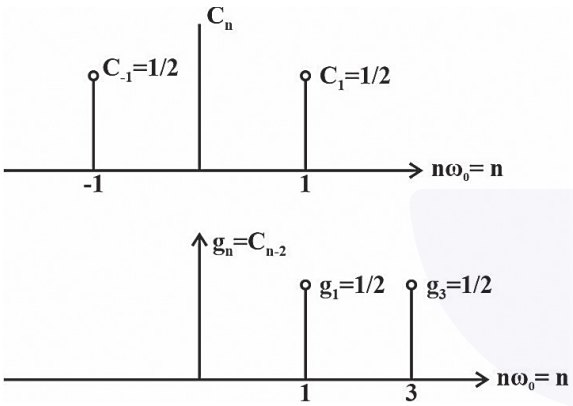


Fig. 3.17

Differentiation property:

$\omega_0 : x(t) \longleftrightarrow C_n : C_n vs\ n\omega_0$

$\omega_0 : g(t) = \frac{dx(t)}{dt} \longleftrightarrow g_n$

$= (jn\omega_0) C_n : g_n\ vs\ n\omega_0$

$\frac{d^2x(t)}{dt^2} \longleftrightarrow (jn\omega_0)^2 C_n$

$\frac{d^3x(t)}{dt^3} \longleftrightarrow (jn\omega_0)^3 C_n$

Integration property:

$\omega_0 : x(t) \longleftrightarrow C_n : C_n vs\ n\omega_0$

$\omega_0 : g(t) = \int_{-\infty}^{t} x(t)dt \longleftrightarrow g_n = \frac{C_n}{jn\omega_0} : g_n\ vs\ n\omega_0$

Periodic convolution:

$x_1(t) \to T_0$

$x_2(t) \to T_0$

$\int_{T_0} x_1(t)x_2(t-\varepsilon)d\varepsilon$: Periodic convolution

$\omega_0 : x_1(t) \longleftrightarrow C_n : C_n vs\ n\omega_0$

$\omega_0 : x_2(t) \longleftrightarrow d_n : d_n vs\ n\omega_0$

$\omega_1 : g(t) x_1(t) * x_2(t) \longleftrightarrow g_n$

$= T_0 C_n d_n : g_n vs\ n\omega_0$

Note: $\mathbf{x_1(t), x_2(t)}$ must have same $\mathbf{T_0}$

11. $g(t) = \sin t * \cos t \longleftrightarrow g_n = ?$

Solution: $\omega_0 = 1: \quad \sin t \longleftrightarrow C_n = \begin{cases} \frac{1}{2i} & : n = 1 \\ \frac{-1}{2i} & : n = -1 \end{cases}$

$\omega_0 = 1: \quad \cos t \longleftrightarrow d_n = \begin{cases} \frac{1}{2} & : n = 1 \\ \frac{1}{2} & : n = -1 \end{cases}$

$g(t) = \sin t * \cos t \longleftrightarrow g_n = T_0 C_n d_n$

$g_n = 2\pi C_n d_n$

$$= \begin{cases} -j\pi/2 & n = 1 \\ j\pi/2 & : n = -1 \end{cases}$$

Multiplication in time domain:

$\omega_0 : x_1(t) \longleftrightarrow C_n : C_n vs\ n\omega_0$

$\omega_0 : x_2(t) \longleftrightarrow d_n : d_n vs\ n\omega_0$

$\omega_1 : g(t) = x_1(t)x_2(t) \longleftrightarrow g_n$

$= C_n * d_n : g_n vs\ n\omega_0$

Note: Valid if $\mathbf{x_1(t)}$ and $\mathbf{x_2(t)}$ have same $\mathbf{T_0}$

12. $g(t) = \cos t \times \cos t$

$\omega_0 = 1: \quad x_1(t)\cos t \longleftrightarrow C_n$

$$= \begin{cases} \frac{1}{2} & : n = 1 \\ \frac{1}{2} & : n = -1 \end{cases} = \left\{\frac{1}{2}, \underset{\uparrow}{0}, \frac{1}{2}\right\}$$

$\omega_0 = 1: \quad x_2(t)\cos t \longleftrightarrow d_n$

$$= \begin{cases} \frac{1}{2} & : n = 1 \\ \frac{1}{2} & : n = -1 \end{cases} = \left\{\frac{1}{2}, \underset{\uparrow}{0}, \frac{1}{2}\right\}$$

$g(t) = x_1(t)x_2(t) \longleftrightarrow g_n$

$= C_n * d_n : g_n vs\ n\omega_0$

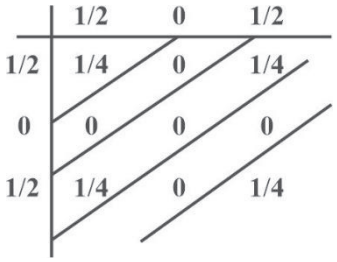


Fig. 3.18

$$g_n = \left\{\frac{1}{4}, 0, \underset{\uparrow}{\frac{1}{2}}, 0, \frac{1}{4}\right\} = \begin{cases} \frac{1}{4} : n = \pm 2 \\ \frac{1}{2} : n = 0 \end{cases}$$

Type: $\cos^2 t = \frac{1}{2} + \frac{1}{4}e^{j2t} + \frac{1}{4}e^{-j2t}$

$$= \frac{1}{2} + \frac{\cos 2t}{2}$$

$\omega_0 = 2$:

$C_0 = 1/2$

$C_1 = 1/4$

$C_{-1} = 1/4$

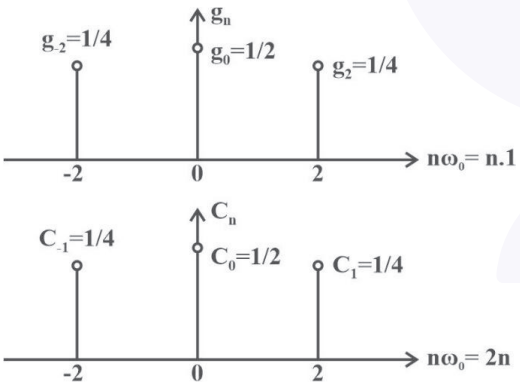


Fig. 3.19

Type 5:

Symmetries in periodic graph:

1. Even symmetry: Even in $\left(-\frac{T_0^+}{2}, \frac{T_0^+}{2}\right)$
2. Odd symmetry: Even in $\left(-\frac{T_0^+}{2}, \frac{T_0^+}{2}\right)$
3.

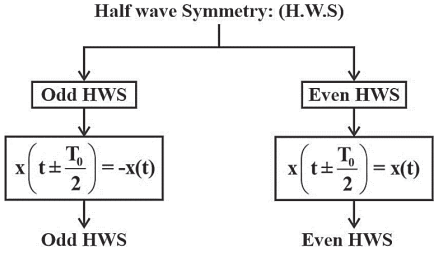


Fig. 3.20

Default HWS: Only odd HWS $x\left(t \pm \frac{T_0}{2}\right) = -x(t)$

Procedure to check HWS:

S.1→ Identify T_0 and track graph from $-\frac{T_0}{2}, \frac{T_0}{2}$.

S.2→ $C_0 = \frac{1}{T_0}\int_{<T_0>} x(t)dt = \frac{Area\ in\ 1T_0}{T_0}$

S.3→

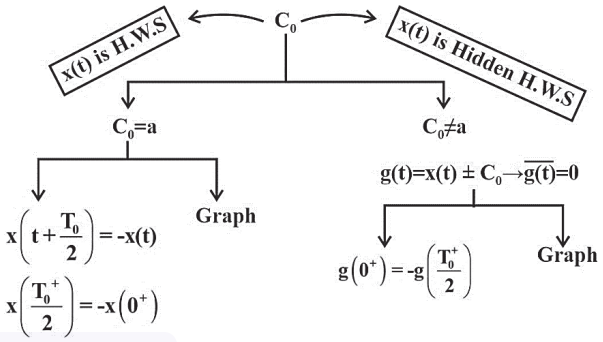


Fig. 3.21

12. Identify HWS

a)

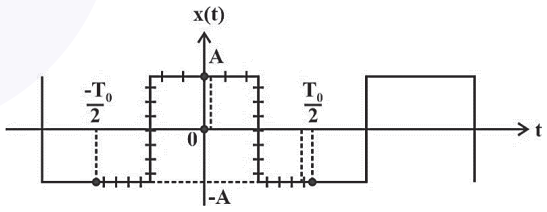


Fig. 3.22

$\rightarrow C_0 = 0$

$x(0^+) = A$

$x\left(\frac{T_0^+}{2}\right) = -A$

$x(0^+) = -x\left(\frac{T_0^+}{2}\right) \rightarrow$ HWS

b)

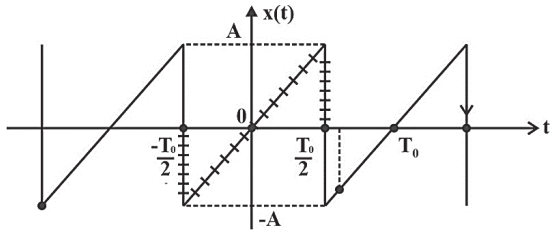


Fig. 3.23

$\rightarrow C_0 = 0$

$x\left(0^+\right) = \in$

$x\left(\frac{T_0^+}{2}\right) = -A + \in$

$x\left(0^+\right) \neq -x\left(\frac{T_0^+}{2}\right) \rightarrow$ No HWS

c)

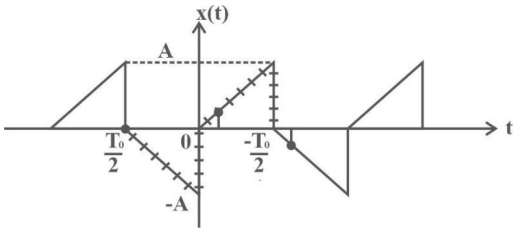


Fig. 3.24

$\rightarrow C_0 = 0$

$x\left(0^+\right) = \in$

$x\left(\frac{T_0^+}{2}\right) = -\in$

$x\left(0^+\right) = -x\left(\frac{T_0^+}{2}\right) \rightarrow$ HWS

d)

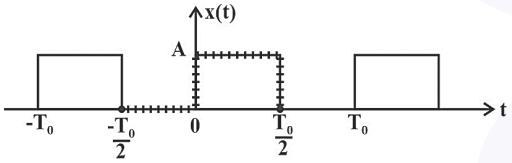


Fig. 3.25

$\rightarrow C_0 = \frac{AT_0/2}{T_0} = A/2 \neq 0$

$\rightarrow g(t) = x(t) - A/2$

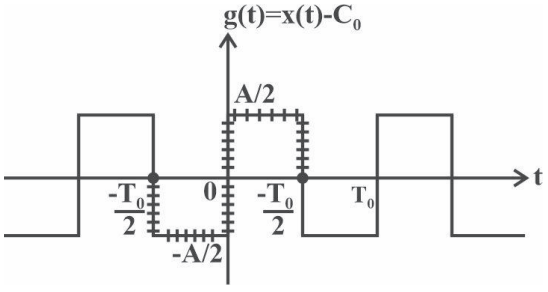


Fig. 3.26

$\rightarrow g_0 = 0$

$\rightarrow g\left(0^+\right) = -g\left(\frac{T_0^+}{2}\right) \rightarrow g(t)$ is HWS $\rightarrow$

x(t) is hidden HWS.

e)

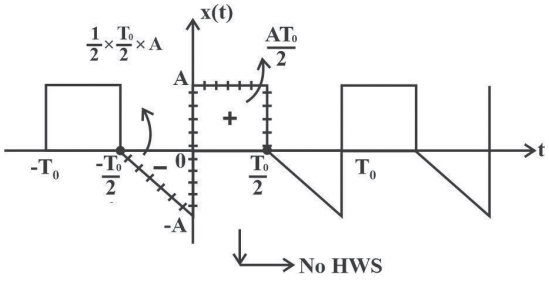


Fig. 3.27

$\rightarrow C_0 = \frac{A}{4} = \frac{A}{2} - \frac{A}{4} = \frac{A}{4}$

$\rightarrow g(t) = x(t) - \frac{A}{4}$

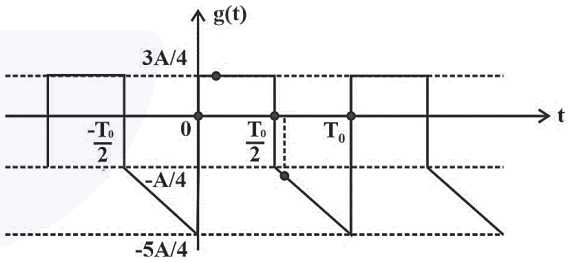


Fig. 3.28

$\rightarrow g_0 = 0$

$\rightarrow g\left(0^+\right) = 3A/4$

$g\left(\frac{T_0^+}{2}\right) = \frac{-A}{4} - \in$

$g\left(0^+\right) \neq -g\left(\frac{T_0^+}{2}\right)$

→ g(t) is not HWS.

→ x(t) is not HWS.

Effect of symmetry on TFSC (a_0, a_n, b_n):

Case 1: $x(t)$: EVEN

$a_0 = \begin{matrix}\nearrow = 0 \\ \searrow \neq 0\end{matrix}$

$a_n \neq 0 \rightarrow$ "a_n will be nonzero for at least 1 value of n."

$b_n = 0$

TFS: $x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$

Harmonics:

1. dc value may or may not be present
2. Harmonics of cosine is decided by a_n
3. All harmonics of sine will be absent

Frequency:

OHz→ decided by a_0

Other frequency → decided by a_n

Case 2: $x(t)$: *ODD*

$$\begin{cases} a_0 = 0 \\ a_n = 0 \\ b_n \neq 0 \rightarrow b_n \text{will be nonzero for at least one value of n.} \end{cases}$$

TFS: $x(t) = \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$

Harmonics:

1. DC will be absent
2. All harmonics of cosine will be absent
3. Harmonics of sine will be decided by b_n

Frequency:

OHz: absent

Other frequency: decided by b_n

Case 3: x(t) is HWS $a_0 = 0$

$$\left.\begin{matrix} a_n = 0 \\ b_n = 0 \end{matrix}\right\} n : EVEN$$

$$\left.\begin{matrix} a_n \neq 0 \\ b_n \neq 0 \end{matrix}\right\} n : ODD$$

TFSC: $x(t) = \sum_{n=1}^{\infty} \left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right)$

$$= \sum_{k=1}^{\infty} a_{(2k-1)} \cos(2k-1)\omega_0 t + b_{(2k-1)} \sin(2k-1)\omega_0 t$$

Harmonics:

1. DC will be absent
2. All even harmonics of sine and cosine both will be absent.
3. All odd harmonics of sine and cosine both will be present.

Frequencies: $f_0, 3f_0, 5f_0, 7f_0 \cdots\cdots,$

Case 4: $x(t) : EVEN + HWS$

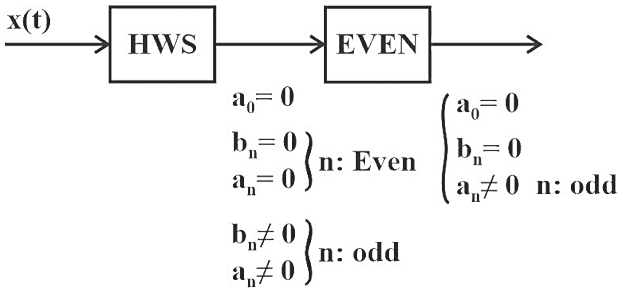


Fig. 3.29

TFS: $x(t) = \sum_{k=1}^{\infty} a_{(2k-1)} \cos(2k-1)\omega_0 t$

Harmonics:

1. DC absent
2. All harmonics of sine absent
3. All even harmonics of cosine absent
4. All odd harmonics of cosine present.

Frequency: $f_0, 3f_0, 5f_0, 7f_0 \cdots\cdots$

Case 5: $x(t) = ODD + HWS.$

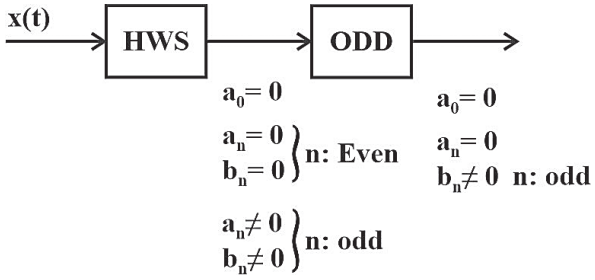


Fig. 3.30

TFS: $x(t) = \sum_{k=1}^{\infty} b_{(2k-1)} \sin(2k-1)\omega_0 t$

Harmonics:

1. DC absent
2. All harmonics of cosine absent
3. All even harmonics of sine absent
4. All odd harmonics of sine present.

Frequency: $f_0, 3f_0, 5f_0, 7f_0 \cdots\cdots$

Quarter wave symmetry:

$EVEN + HWS \xleftrightarrow{T_0/4 \text{ shift}} ODD + HWS$

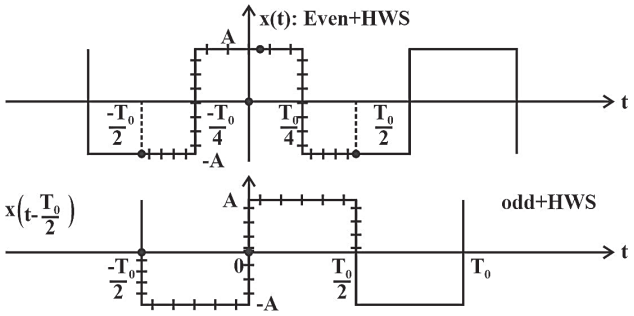


Fig. 3.31

Effect of addition or subtraction of DC value on FSC:

$$x(t) = a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + \cdots$$
$$+ b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + \cdots$$

$$x(t) \xleftrightarrow{F.S.C.} \begin{cases} a_0 : & n = 0 \\ a_n : & n \geq 1 \\ b_n : & n \geq 1 \end{cases}$$

$$g(t) = x(t) \pm A = (a_0 \pm A) + a_1 \cos \omega_0 t$$
$$+ a_2 \cos 2\omega_0 t + \cdots$$
$$+ b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + \cdots$$

$$g(t) = x(t) \pm A \xleftrightarrow{F.S.C.} \begin{cases} g_0 = a_0 \pm A & n = 0 \\ a_n : n \geq 1 \\ b_n : n \geq 1 \end{cases}$$

Note: Upward or downward shifting of graph does not affect the harmonics present in it, it will affect the dc value of x(t).

Procedure to calculate harmonics:

S.1→ Identify T_0 and track $\left(\frac{-T_0}{2}, \frac{T_0}{2}\right)$

S.2→ Calculate C_0

S.3→

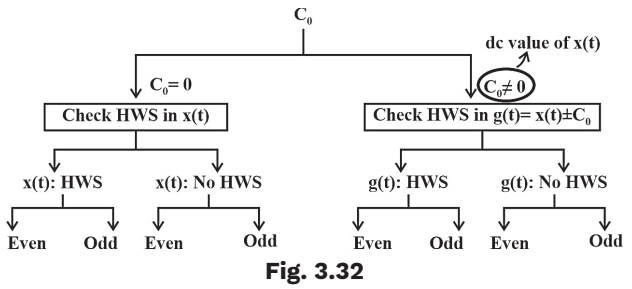
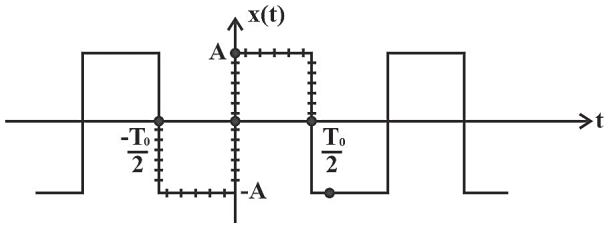


Fig. 3.32

Asymmetric square wave:



1. $C_0 = 0$
2. HWS in x(t) = ? →HWS in x(t)
3. Odd

$x(t)$: $HWS + ODD \rightarrow$ Case (5)

Symmetric square wave:

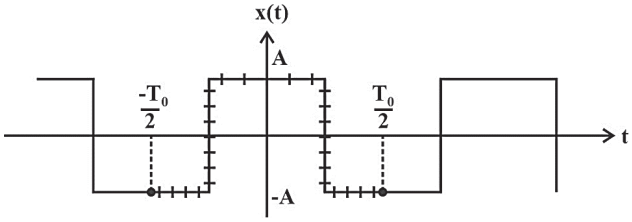


Fig. 3.33

1. $C_0 = 0$
2. HWS in x(t) = ? **Yes**
3. x(t) : Even

$EVEN + HWS \rightarrow$ Case (4)

Asymmetric triangle wave:

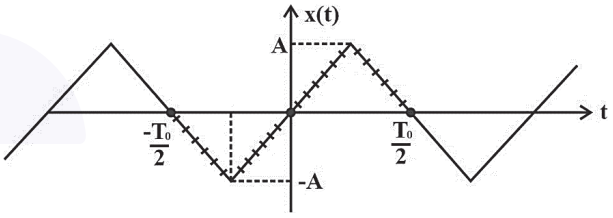


Fig. 3.34

1. $C_0 = 0$
2. HWS in x(t) = Yes
3. x(t) : Even

$ODD + HWS \rightarrow$ Case (5)

Symmetric triangle wave:

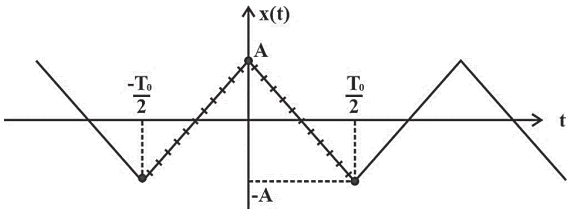


Fig. 3.35

1. $C_0 = 0$

2. HWS in x(t) = Yes
3. x(t) : Even

Even + HWS → Case (4)

Note: Sinusoidal + Periodic → Single discrete non-sinusoidal + Periodic→ Infinite no. of discrete frequency.

Type 4:

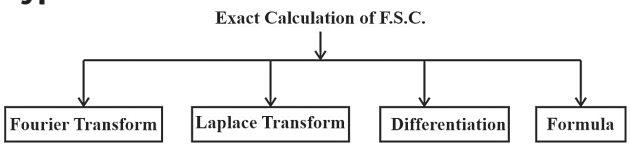


Fig. 3.35

Relation:

1. $x(t) \xleftrightarrow{L.T} x(s)$

   $x(t) \xleftrightarrow{F.T} x(\omega)$

2.

$x(t) \longleftrightarrow X(s)$	$x(t) \longleftrightarrow X(\omega)$
$x(t-t_0) \longleftrightarrow$	$x(t-t_0) \longleftrightarrow$
$e^{-st_0}X(s)$	$e^{-j\omega t_0}X(\omega)$
$x(t+t_0) \longleftrightarrow$	$x(t+t_0) \longleftrightarrow$
$e^{st_0}X(s)$	$e^{j\omega t_0}X(s)$

3. Some standard FT

   a) $x(t) = \delta(t) \longleftrightarrow X(\omega) = 1$

   b)

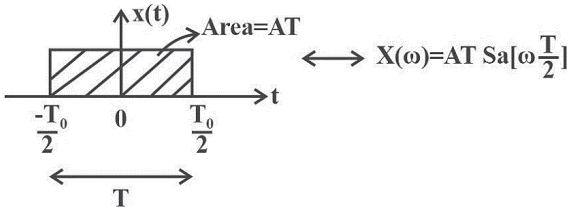


Fig. 3.36

   c)

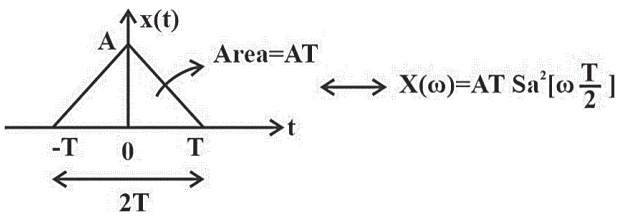


Fig. 3.37

4. Some standard LT $\delta(t) \longleftrightarrow 1$

$$u(t) \longleftrightarrow \frac{1}{s}$$

$$r(t) = tu(t) \longleftrightarrow \frac{1}{s^2}$$

$$t^n u(t) \longleftrightarrow \frac{n!}{s^{n+1}}$$

$$\sin \omega_0 t\, u(t) \longleftrightarrow \frac{\omega_0}{s^2 + \omega_0^2}$$

$$\cos \omega_0 t + u(t) \longleftrightarrow \frac{s}{s^2 + \omega_0^2}$$

Procedure:

S.1→ Identify T_0 and track anyone time period.

S.2→ $C_0 = a_0 = r_0 = \frac{1}{T_0}\int_{T_0} x(t)dt = \frac{Area\ in\ 1T_0}{T_0}$

S.3→ Extract 1 T_0 of $x(t) = x_{T_0}(t)$

S.4→

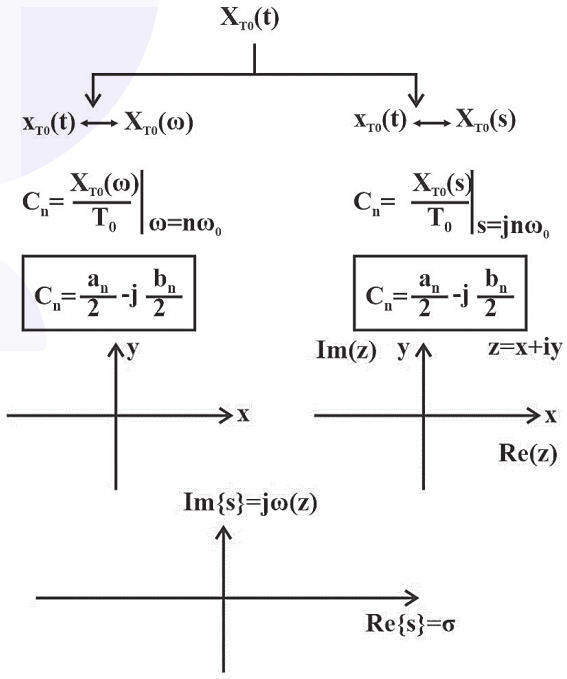


Fig. 3.38

Real $\{s\} = -\infty < \sigma < \infty$

$-\infty <$ Real $\{s\} < \infty$

12.

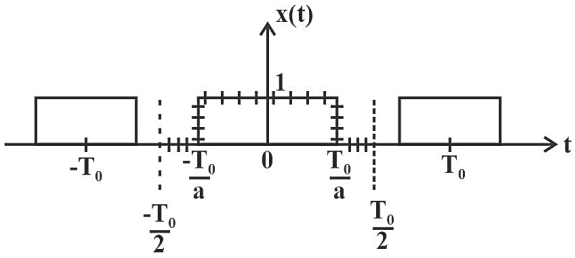


Fig. 3.39

1. $C_0 = \frac{2T_0}{aT_0} = \frac{2}{a}$

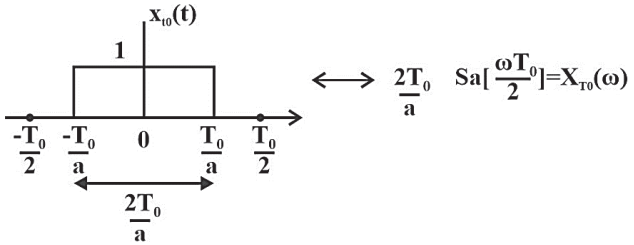


Fig. 3.40

$$C_n = \frac{X_{T_0}(\omega)}{T_0}\bigg|_{\omega=n\omega_0} = \frac{2T_0}{aT_0} Sa\left[\frac{\eta\omega_0 T_0}{a}\right]$$

$$= \frac{2}{a} Sa\left[\frac{n2\pi}{a}\right]$$

$$\boxed{\omega_0 T_0 = 2\pi}$$

$$\rightarrow C_n = \begin{cases} \frac{2}{a} & n = 0 \\ \frac{2}{a} Sa\left(\frac{2n\pi}{a}\right) & n \neq 0 \end{cases}$$

$$\rightarrow \begin{cases} a_0 = 2/a \\ a_n = \frac{4}{a} Sa\left(\frac{2n\pi}{a}\right) \\ b_n = 0 \end{cases}$$

Calculation:

1. Frequencies present in x(t) = ?

$C_n = 0$

$$\frac{2}{a} Sa = \left(\frac{2n\pi}{a}\right) = 0$$

$$\frac{2n\pi}{a} = K\pi \quad K \in I, K \neq 0$$

$n = \frac{Ka}{2} : K \in I, K \neq 0 \rightarrow$ Absent harmonics

Note:

$Sa(t) = 0$

$t = K\pi \quad K \in I$

$K \neq 0$

2.

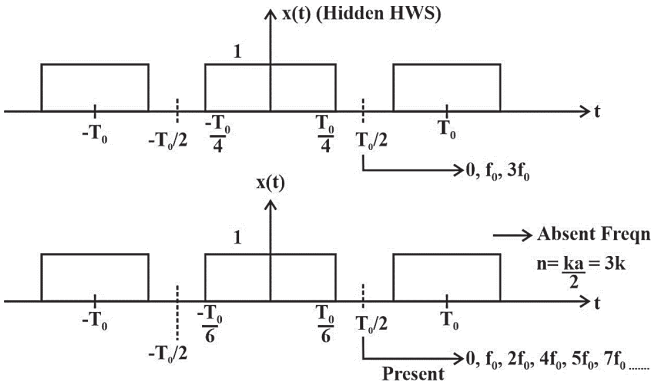


Fig. 3.41

Example 13. Frequencies present in x(t) = ?

Solution: Absent: $C_n = 0$

$$\frac{1}{a}\left[Sa\left(\frac{n\pi}{a}\right)\right]^2 = 0$$

$$\frac{n\pi}{a} = K\pi \quad K \in J, \; K \neq 0$$

$n = Ka : K \in I \quad K \neq 0$

Points:

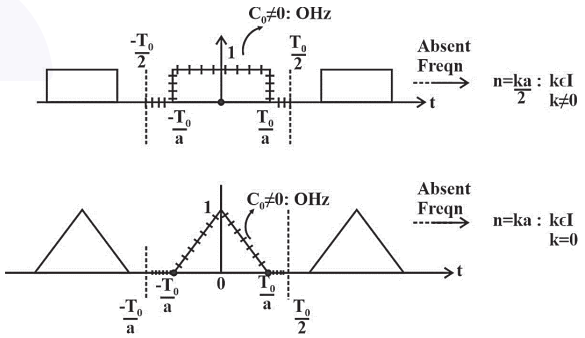


Fig. 3.42

Laplace transform:

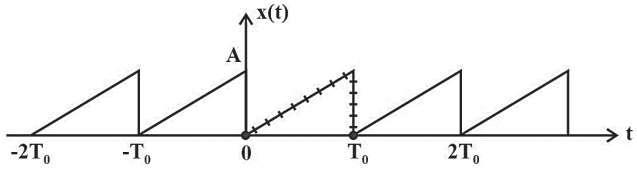


Fig. 3.43

1. $C_0 = \frac{A}{2} \rightarrow OHz$

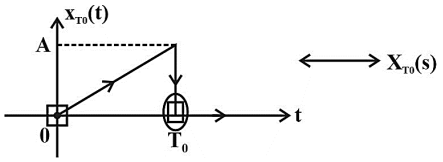


Fig. 3.44

$$X_{T_0}(t) = \frac{A}{T_0} r(t) - Au(t - T_0) - A / T_0 r(t - T_0)$$

$$X_{T_0}(s) = \frac{A}{T_0 s^2} - \frac{Ae^{-sT_0}}{s} - \frac{Ae^{-sT_0}}{T_0 s^2}$$

$$C_n = \left\{\frac{X(s)}{T_0}\right\}_{s=jn\omega_0}$$

$$C_n = \frac{1}{T_0}\left\{\frac{-A}{T_0 n^2 \omega_0^2} - \frac{Ae^{-jn\omega_0 T_0}}{jn\omega_0} + \frac{Ae^{-jn\omega_0 T_0}}{T_0 n^2 \omega_0^2}\right\}$$

$\omega_0 T_0 = 2\pi$

$e^{-jn\omega_0 T_0} = 1$

$$C_n = \frac{-Ae^{-jn2\pi}}{jn\omega_0 T_0} = \frac{jA}{2n\pi}$$

$$C_n = \begin{cases} A/2: & n = 0 \\ \dfrac{jA}{2n\pi}: & n \neq 0 \end{cases}$$

14. Frequency present in x(t)

$C_n \neq 0 \quad \forall \quad n \in I$

$f = 0Hz, f_0, 2f_0, 3f_0, 4f_0 \ldots..$

15.

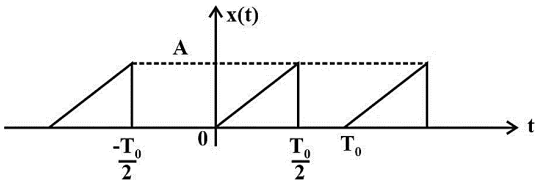


Fig. 3.45

$\rightarrow C_0 = \frac{A}{4}$

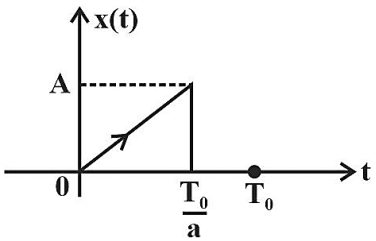


Fig. 3.46

$$x_{T_0}(t) = \frac{2A}{T_0} r(t) - Au\left(t - \frac{T_0}{2}\right) - \frac{2A}{T_0} r\left(t - \frac{t_0}{2}\right)$$

$$X_{T_0}(s) = \frac{2A}{T_0 s^2} - \frac{Ae^{-sT_0/2}}{s} - \frac{2A}{T_0}\frac{e^{-sT_0/2}}{s^2}$$

$e^{-jn\pi} = (-1)^n$

$$C_n = \left\{\frac{X_{T_0}(s)}{T_0}\right\}_{S=jn\omega_0} = \frac{-2A}{T_0^2 n^2 \omega_0 2} - \frac{Ae^{-\frac{jn\omega_0 T_0}{2}}}{jn\omega_0 T_0} + \frac{2A}{T_0^2}\frac{e^{-\frac{jn\omega_0 T}{2}}}{n^2\omega_0 2}$$

$$= \frac{2A}{T_0^2 n^2 \omega_0^2}\left[(-1)^n - 1\right] + \frac{jA(-1)^n}{n\omega_0 T_0}$$

$$= \frac{2A}{n^2 4\pi^2}\left[(-1)^n - 1\right] + \frac{jA(-1)^n}{2n\pi}$$

Differentiation method:

S.1→ Track one time period: ω_0, T_0, C_0

S.2→ Differentiate the signal until.

$$\frac{d^n x(t)}{dt^n} = Ag(t - t_0)$$

$$\frac{d^n x(t)}{dt^n} = Ag(t - T_0) + Bx(t)$$

$$\frac{d^n x(t)}{dt^n} = Ag(t - t_0) + B$$

g(t) is impulse train of unit area impulses

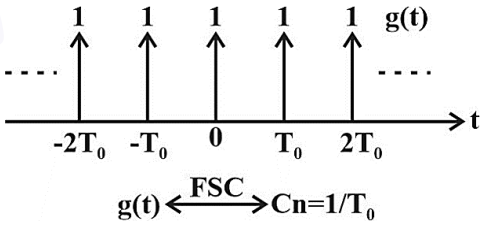


Fig. 3.47

S.3→ $FSC\left[\frac{d^n x(t)}{dt^n}\right] = (jn\omega_0)^n C_n$

Note:

1. While differentiating x(t), if addⁿ or subⁿ of dc is observed, it must be ignored.
2. Always take $t=T+0^+$ interval while differentiating the signal.

16.

1. $C_0 = 0$
   T_0, ω_0

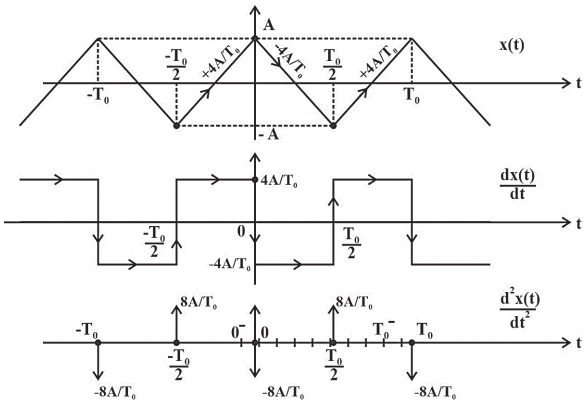


Fig. 3.48

$$\frac{d^2x(t)}{dt^2} = -\frac{8A}{T_0}\delta(t) + \frac{8A}{T_0}\delta\left(t - \frac{T_0}{2}\right)$$

$$\frac{d^2x(t)}{dt^2} = -\frac{8A}{T_0}g(t) + \frac{8A}{T_0}g\left(t - \frac{T_0}{2}\right)$$

$$FSC\left[\frac{d^2x(t)}{dt}\right] = FSC\left[\frac{-8A}{T_0}g(t)\right] + FSC\left[\frac{8A}{T_0}g\left(t - \frac{T_0}{2}\right)\right]$$

$$\left(jn\omega_0\right)^2 C_n = -\frac{8A}{T_0}\times\frac{1}{T_0} + \frac{8A}{T_0}\times\frac{1}{T_0}e^{\frac{-jn\omega_0 T_0}{2}}$$

$$\omega_0 T_0 = 2\pi$$

$$\left(jn\omega_0\right)^2 C_n = \frac{8A}{T_0^2}\left[-1 + e^{-jn\pi}\right]$$

$$\left(jn\omega_0\right)^2 = -n^2\omega_0^2$$

$$C_n = \frac{8A}{n^2\omega_0^2T_0^2}\left[1 - e^{-jn\pi}\right]$$

$$C_n = \frac{8A}{n^2 4\pi^2}\left[1 - (-1)^n\right]$$

$$C_n = \frac{2A}{n^2\pi^2}\left[1 - (-1)^n\right]$$

$$C_n = \begin{cases} 0 & n = 0 \\ \dfrac{2A}{n^2\pi^2}\left(1 - (-1)^n\right] & n \neq 0 \end{cases}$$

Important conclusions:

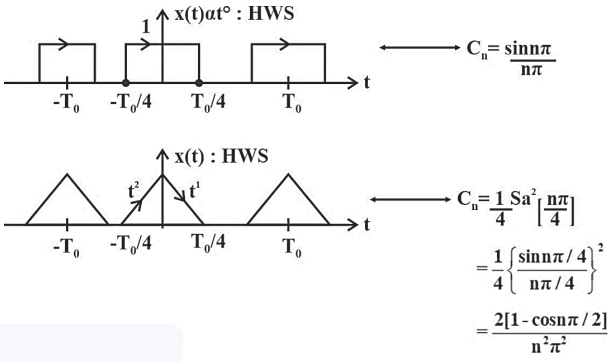
1. **If F.T. of 1 T_0 is known** → use F.T. methods.
2. **FWR versions (Full wave rectifier)**
   → Laplace methods
   → Differentiation method if differentiation generates impulses
3. **HWR version (Half wave rectifier)**
   → Laplace method
   → Don't use differentiation method.
4. If $(a_n)_{n=k}$ or $(b_n)_{n=k}$ is to be calculated.
   Use formula directly.
5. 

Fig. 3.49

If x(t) is HWS + x(t) is having similar type of variation.

Important:

$$x(t)\,\alpha\, t^k \leftrightarrow |C_n|\ \alpha\ \frac{1}{n^{k+1}}$$

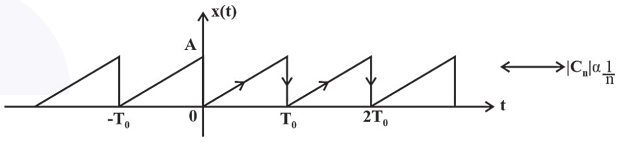


Fig. 3.50

1. x(t): No HWS
2. x(t): No similar variation

   $|C_n|\,\alpha\,\frac{1}{n^2}$ → Can use it

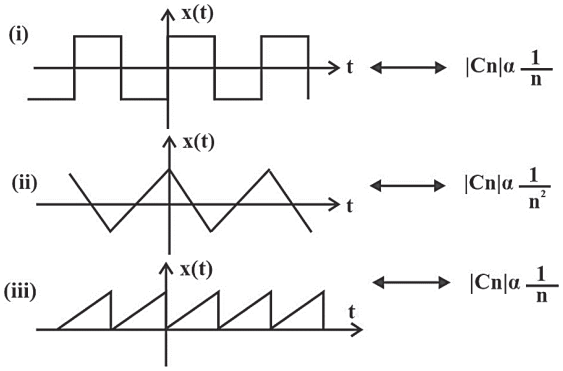


Fig. 3.51

17.

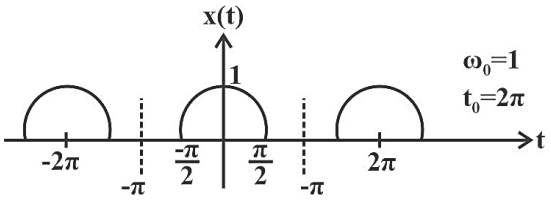


Fig. 3.52

$$x(t)=\cos t:\frac{-\pi}{2}\le t\le \pi/2$$

$$a_1=?$$

$$b_1=?$$

$$b_1=0$$

$$a_n=\frac{2}{T_0}\int_{-\pi}^{\pi}x(t)\cos n\omega_0 t\,dt$$

$$a_1=\frac{2}{2\pi}\int_{-\pi/2}^{\pi/2}\cos t\cos t\,dt$$

$$a_1=\frac{1}{\pi}x\int_{-\pi/2}^{\pi/2}\left(\frac{1}{2}+\frac{\cos 2t}{2}\right)dt=\frac{1}{2}$$

Type 7: Problems based on power calculation:

x(t) is power signal, which is periodic with period = T_0

Time domain power: $P_x=\frac{1}{T_0}\int_{T_0}|x(t)|^2\,dt\rightarrow$ Total or absolute power.

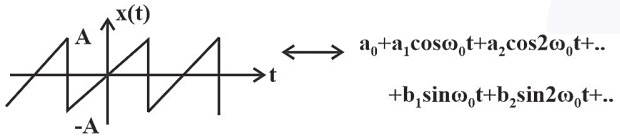


Fig. 3.53

$$P_x=\frac{1}{T_0}\int_{T_0}|x(t)|^2\,dt\quad P_x=a_0^2+\sum_{n=1}^{\infty}\left\{\frac{a_n^2}{2}+\frac{b_n^2}{2}\right\}$$

Parseval theorem:

$$P_x=\frac{1}{T_0}\int_{T_0}|x(t)|^2\,dt=a_0^2+\sum_{n=1}^{\infty}\left\{\frac{a_n^2}{2}+\frac{b_n^2}{2}\right\}$$

$$a_0^2+\sum_{n=1}^{N}\left\{\frac{a_n^2}{2}+\frac{b_n^2}{2}\right\}=m\%\left\{\frac{1}{T_0}\int_{T_0}|x(t)|^2\,dt\right\}$$

$$\begin{cases}C_n=\frac{a_n}{2}-j\frac{b_n}{2}\\ C_{-n}=\frac{a_{-n}}{2}-j\frac{b_{-n}}{2}=\frac{a_n}{2}+j\frac{b_n}{2}\\ |C_n|^2+|C_{-n}|^2=\frac{a_n^2}{2}+\frac{b_n^2}{2}\end{cases}$$

$$|C_n|=\frac{\sqrt{a_n^2+b_n^2}}{2}$$

$$P_x=a_0^2+\sum_{n=1}^{\infty}\frac{a_n^2}{2}+\frac{b_n^2}{2}$$

$$=a_0^2+\sum_{n=1}^{\infty}|C_n|^2+|C_{-n}|^2$$

$$=C_0^2+|C_1|^2+|C_{-2}|^2+|C_2|^2+|C_{-2}|^2+\cdots$$

$$P_x=\frac{1}{T_0}\int_{T_0}|x(t)|^2\,dt=a_0^2+\sum_{n=1}^{\infty}\frac{a_n^2}{2}+\frac{b_n^2}{2}$$

$$=C_0^2+\sum_{\substack{n=-\infty\\ n\neq 0}}^{\infty}|C_n|^2$$

Case 1: $x(t)$: Real: $\rightarrow \begin{matrix}a_0\\ a_n\\ b_n\end{matrix}\rightarrow$ Real:

$$x(t)=a_0+\sum_{n=1}^{\infty}a_n\cos n\omega_0 t+b_n\sin n\omega_0 t$$

$$P_x=a_0^2+\sum_{n=1}^{\infty}\frac{a_n^2}{2}+\frac{b_n^2}{2}=C_0^2+\sum_{\substack{n=-\infty\\ n\neq 0}}^{\infty}|C_n|^2=\sum_{n=-\infty}^{\infty}|C_n|^2$$

Case 2:

$x(t)$: complex or purely $\rightarrow \begin{matrix}a_0\\ a_n\\ b_n\end{matrix}\rightarrow$ Complex or I

$$x(t)=a_0+\sum_{n=1}^{\infty}a_n\cos n\omega_0 t+b_n\sin n\omega_0 t$$

$$P_x=|a_0|^2+\sum_{n=1}^{\infty}\left\{\frac{|a_n|^2}{2}+\frac{|b_n|^2}{2}\right\}$$

$$=|C_0|^2+\sum_{\substack{n=-\infty\\ n\neq 0}}^{\infty}|C_n|^2=\sum_{n=-\infty}^{\infty}|C_n|^2$$

Case 3: x(t): Real or complex or I.

$$P_x = \frac{1}{T_0}\int_{T_0} |x(t)|^2\, dt = |a_0|^2$$

$$+\sum_{n=1}^{\infty}\left\{\frac{|a_n|^2}{2}+\frac{|b_n|^2}{2}\right\} = \sum_{n=-\infty}^{\infty}|C_n|^2$$

Important conclusion:

Power of kth harmonic:

$$= \left\{\frac{|a_K|^2+|b_K|^2}{2}\right\} = |C_K|^2 + |C_{-K}|^2$$

$\downarrow x(t)$ Real: $|C_K| = |C_{-K}|$

$$\frac{a_k^{\,2}+b_k^{\,2}}{2} = \frac{r_k^{\,2}}{2} = 2|C_k|^2$$

18.

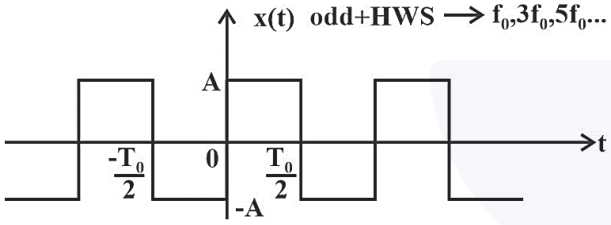


Fig. 3.54

$$|C_n|^2 \alpha \frac{1}{n}$$

$$|C_n|^2 \alpha \frac{1}{n^2}$$

1. $\frac{P_{II}}{P_{III}} = \frac{0}{P_{III}} = 0$

2. $\frac{P_{III}}{P_{IV}} = \frac{|C_3|^2}{|C_5|^2} = \frac{25}{9}$

Dirichlet conditions:

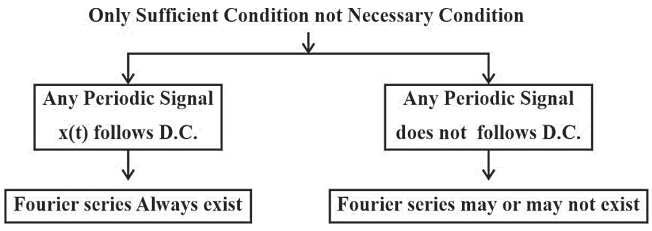


Fig. 3.55

Condition 1: $\int_{T_0} |x(t)|\, dt =$ *Finite*

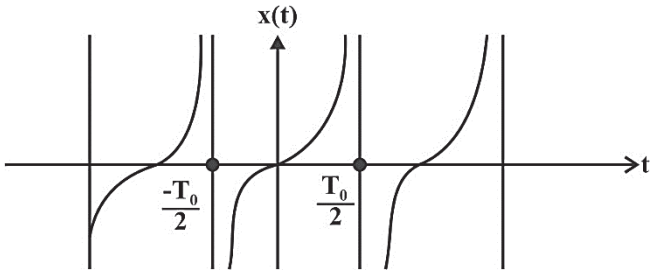


Fig. 3.56

Condition 2: x(t) must have finite no. of maxima minima in one of its time period.

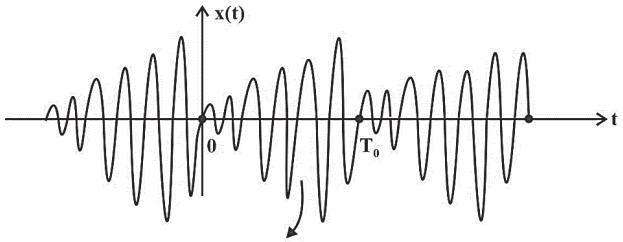


Fig. 3.57

Condition 3: x(t) must have

Finite number of finite size discontinuity, in one of its time period.

1.

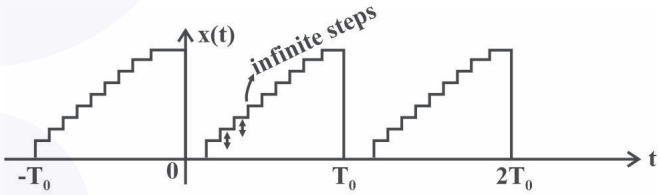


Fig. 3.58

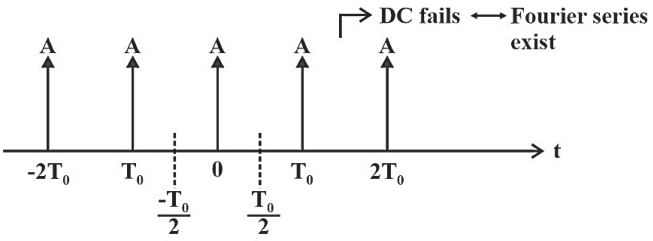


Fig. 3.59

Gibb's phenomenon:

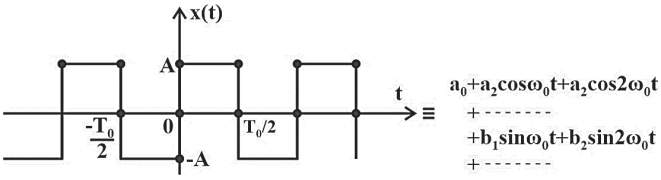


Fig. 3.60

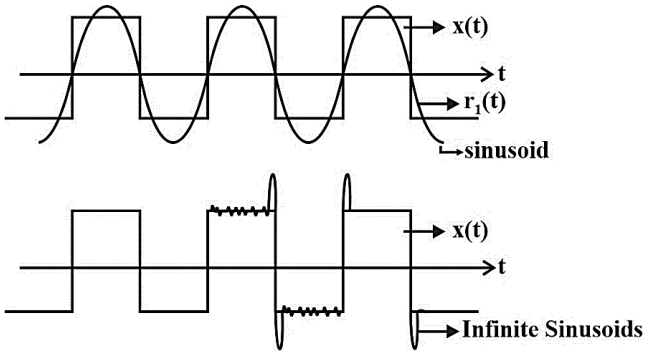


Fig. 3.61

Statement 1: Any non-sinusoidal periodic signal can always be exactly written as the weighted sum of mutually orthogonal, harmonically related sinusoids is not completely true.

Statement 2: The statement that Fourier series converges at each and every point of the time domain signal is not completely true.

Level 1 Exercise

1. The Fourier series representations of a periodic current

$[2+6\sqrt{2}\cos\omega t+\sqrt{48}\sin 2\omega t]A$

The effective value of the current is

(A) $[2+6+\sqrt{24}]A$ (B) 8A

(C) 6A (D) 2A

2. Match List-1 (Properties) with List-II (Characteristics of the trigonometric form) in regard to Fourier series of periodic f(t) and select the correct answer using the codes given below the lists:

List I	**List II**
A. $f(t)+f(-t)=0$	1. Even harmonics can exist
B. $f(t)-f(-t)=0$	2. Odd harmonics can exist
C. $f(t)+f(t-T/2)=0$	3. The dc and cosine terms can exist
D. $f(t)-f(t-T/2)=0$	4. sine terms can exist
	5. cosine terms of even harmonics can exist

Codes:

	A	B	C	D
(A)	4	5	3	1
(B)	3	4	1	2
(C)	5	4	2	3
(D)	4	3	2	1

Level 2 Exercise

1. Assertion (A): A periodic function satisfying Dirichlet conditions can be expanded into Fourier series.

Reason (*R*) **:** A periodic function can be reconstructed from $\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos n\omega_0 t+\sum_{n=1}^{\infty}b_n\sin n\omega_0 t$ for very large n, excluding infinity.

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

2. Assertion (A): In the exponential Fourier representation of a real-valued periodic function f(t) of frequency f_0 the coefficients of the terms $e^{j2\pi nf_0t}$ and $e^{-j2\pi nf_0t}$ are negatives of each other.

Reason (R): The discrete magnitude spectrum of f(t) is even and the phase spectrum is odd.

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) Ais true but R is false

(D) Ais false but R is true

3. Match List-I (Nature of periodic function) with List-II (Properties of

spectrum function) and select the correct answer using the codes given below the lists:

List-I

A. Impulse train

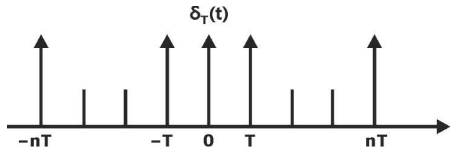


Fig. 3.62

B. Full-wave rectified sine function

C. $2\sin(2\pi t/6)\cos(4\pi t/6)$

D.

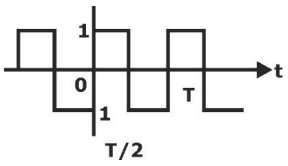


Fig. 3.63

List-II

1. Only even harmonics are present
2. Impulse train with strength 1/T
3. $\alpha_3 = 1/2j; \alpha_{-3} = -1/2j; \alpha_1 = -1/2j; \alpha_{-1} = 1/2j$
4. Only odd harmonics are present
5. Both even and odd harmonics are present

Codes:

	A	B	C	D
(A)	5	2	3	4
(B)	2	1	4	3
(C)	5	2	4	3
(D)	2	1	3	4

Level 1 Solutions

1. Solution: (B)

$$x(t) = 2t6\sqrt{2}\omega put + \sqrt{HB}.\sin(2\omega t)$$

$$RMS = \sqrt{MSV}$$

since, $\omega_1 = \omega_1 ur = 2\omega$

Individual ω sinusoids are of different frequencies and are orthogonal to each other.

$$MSV = H + \frac{36\times 2}{2} + \frac{HB}{2}$$

$$= H + 36 + 2H$$

$$MSV = 6H$$

$$RMS = \sqrt{6H} = 8\,A.$$

Option 'B' is correct.

2. Solution: (D)

$$(A)\, f(t) + f(t) = 0$$

$$f(t) = -f(t)$$

$f(t)$ is odd signal and hence, dc term and cosine term... Exist. Only sine terms exist.

$$A \to 4$$

$$(B)\, f(t) - f(-t) = 0$$

$$f(t) = f(-t)$$

f(t) is oven signal; hence only dc ξ cosine terms will exist.

$$B \to 3$$

$$(c)\, f(t) + f(t - T/2) = 0$$

Condition for HWS; hence, odd harmonics of cosine and sine terms can exist.

$$C \to 2$$

$$(D)\, f(t) - f(t - T/2) = 0$$

$$f(t) = f(t - T/2)$$

Condition for even HWS, even harmonics can exist.

$$(D) \to 1$$

Hence option 'D' is correct.

Level 2 Solutions

1. Solution: (C)

Assertion is true according to dirichlet conditions.

Reason is false because the periodic function cannot be reason $x(t) = a_0 + \sum_{n=1} a_n \cos \omega_0 t + \sum_{n=1} b_n \sin(\omega_s t)$

Hence, option 'c' is correct.

2. Solution: (D)

Assertion is false because for a real-valued periodic function f(t), $C_n = C_{-n}^*$ which is not being regahives of each other.

Reason is true because for a real-valued function, ... magnitude spectrum is even and phase spectrum is odd; hence, option 'D' is correct.

3. Solution: (C)